

DOCUMENT: PROPOSED SOLUTION

Phase	Project Design Phase
Document	Proposed Solution
Project Name	PetPaws – Pet Foods & Accessories Store
Team ID	[TEAM-ID]
Date	14 June 2026
Maximum Marks	2 Marks

Proposed Solution — PetPaws

S.No.	Parameter	Description
1	Problem Statement	Pet owners in India lack a dedicated, domain-specific e-commerce platform for all pet care needs. Existing general platforms (Amazon, Flipkart) do not offer pet-type filtering, breed-specific recommendations, verified pet purchase reviews, or an integrated admin dashboard for pet store management. This results in a fragmented, unreliable, and time-consuming shopping experience for pet parents, and operational inefficiency for pet store administrators.
2	Idea / Solution Description	PetPaws is a full-stack MERN (MongoDB, Express.js, React.js, Node.js) e-commerce web application exclusively dedicated to pet foods and accessories. It provides: (1) A product catalog spanning 15 categories and 8 pet types with advanced keyword search and multi-filter capabilities. (2) A secure JWT-based authentication system for both customers and administrators. (3) A MongoDB-persisted shopping cart and wishlist with real-time badge updates. (4) A 3-step checkout process supporting COD, Card, UPI, and Net Banking payments. (5) A complete order lifecycle management system with visual status timelines and tracking numbers. (6) A product review system with Verified Purchase badge auto-assignment. (7) An admin dashboard with KPI analytics, revenue charts, low-stock alerts, and full CRUD for products, orders, users, and store settings.
3	Novelty / Uniqueness	(a) Domain Exclusivity: Unlike general e-commerce platforms, PetPaws is purpose-built for the pet care domain with 15 pet-specific product categories and 8 pet-type filters that no general platform offers natively. (b) Verified Purchase Trust System: The Verified Purchase badge is auto-assigned using server-side delivery status verification — providing trust that is domain-specific to

		pet owners. (c) Soft Delete with Historical Integrity: Products are never hard-deleted; isActive flags preserve the integrity of order history even after product removal. (d) Dynamic Discount Engine: A pre-save Mongoose hook automatically computes discountedPrice from base price and discount percentage, ensuring price consistency. (e) Admin Revenue Intelligence: An aggregation pipeline-powered monthly revenue chart and top-products ranking system provides actionable business insights without third-party BI tools.
4	Social Impact / Customer Satisfaction	(a) Pet Welfare: By connecting pet owners with quality-verified, appropriately categorised products, PetPaws directly contributes to better pet health and welfare outcomes in India. (b) Informed Purchasing: Verified Purchase reviews and detailed product descriptions empower pet owners to make informed decisions, reducing the risk of purchasing unsuitable or harmful products for their animals. (c) Time Efficiency: The streamlined 3-step checkout, saved addresses, wishlist, and order tracking features save busy pet parents significant time compared to browsing general platforms. (d) SME Empowerment: Independent pet stores gain a professional digital presence and operational analytics through the admin dashboard, levelling the playing field against large retail chains. (e) Accessibility: Responsive mobile-first design ensures the platform is accessible to pet owners across urban and semi-urban India.
5	Business Model (Revenue Model)	(a) Commission-Based Model (Primary): PetPaws charges a 5–10% commission on each transaction processed through the platform from partner pet stores and independent sellers. (b) Premium Listings: Sellers pay a listing fee for featured placement in the Featured Products section on the homepage. (c) Subscription Plans: Admin accounts can be offered on a tiered SaaS subscription model (Basic, Professional, Enterprise) based on product listing volume and analytics depth. (d) Advertising Revenue: Targeted banner advertising space on the homepage carousel can be sold to pet food and accessory brands. (e) Tax Integration: A configurable 5% GST is applied at checkout, with the platform managing compliance reporting for B2B customers.
6	Scalability of the Solution	(a) Architecture: The three-tier MERN architecture (React SPA → Express REST API → MongoDB) is inherently horizontally scalable. Each layer can be scaled independently on cloud infrastructure (Vercel for frontend, Railway/Render for backend, MongoDB Atlas for database). (b) Database: MongoDB's document model and Atlas's auto-sharding support petabyte-scale data without schema migrations. Text indexes on product fields support full-text search at scale. (c) API Design: RESTful API design with pagination (12 products per

		page, 20 orders per page) ensures consistent performance as data volume grows. (d) State Management: React Context API can be migrated to Redux Toolkit for enterprise-scale state management without architectural refactoring. (e) Authentication: JWT-based stateless authentication eliminates server-side session storage, enabling the API to serve millions of concurrent users behind a load balancer.
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